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THEORY OF FLUID AND CRYSTALLIZED INTELLIGENCE:

A CRITICAL EXPERIMENT[1]

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The theory of fluid and crystallized intelligence propounded 20 years ago by the author has since accumulated support. However, the crucial issue of whether 1 or 2 general factors subtend intellectual performances has lacked an experiment adequately designed for accurate, determinate, simple-structure rotation at the 2nd order. By factoring culturally embedded with culture-fair intelligence measures on a background of pure personality primaries (N = 277 7th and 8th grade boys and girls), it is shown that 2 general factors indeed exist. A review, with some mathematical formulations, is given of the theory's implications for the nature-nurture ratio, brain injury, standard deviation of the IQ, growth curves, the concept of a relational difficulty hierarchy, test standardization, and the relative validities of traditional and culture-fair intelligence tests

PRESENT TRENDS IN INTELLIGENCE CONCEPTS

Since Spearman's demonstration of the general ability factor (g) first put intelligence testing in the framework of scientific research, the field has experienced three major theoretical developments, which, however, have not hitherto been interrelated.

1. Thurstone's (1938) proof that any comprehensive array of ability measures will factor uniquely into a set of simple-structure, oblique, primary abilities, from which g can be obtained as a second-order factor, more uniquely determinable than by the first-order approach.

2. The notion, most frequently associated with the name of Piaget (1947) but widely surmised, and stated very clearly by Ferguson (1956), for example (see also the review by Hunt, 1961), that the formal structure of abilities is partly developmentally determined by what might be called "generalized solution instruments" (i.e., habits the acquisition of which becomes a key, opening the way to rapid advance in some general area of cognitive problem solving). These we shall briefly call "aids."

3. The theory of "fluid and crystallized general abilities," which states that with more refined analytical methods the general ability factor now measured by intelligence tests will be found to be not one factor but *two*. In spite of marked cooperativeness of loading pattern (Cattell, 1952), which makes them difficult to separate, they have properties differing in vital ways for educational and clinical prediction.

This article puts forward entirely new evidence specifically on the theory of fluid and crystallized g factors, but integrates all three of the above and has implications for practical testing in terms of culture-fair intelligence tests.

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THEORY OF FLUID AND CRYSTALLIZED GENERAL ABILITY

Although the theory of fluid and crystallized general ability was first stated 20 years ago (Cattell, 1941, 1943), and since developed more precisely (Cattell, 1950, pp. 477–491; Cattell, 1957a, pp. 871–879; Cattell, 1957b), the momentum of custom among intelligence testers, and the absence of sufficiently sustained research on the theory, have hitherto denied to practical test construction and use the advantages in understanding that might be expected to acrue from such dual measurement. The main problem in *research* advance is one of organization, since, as the following summary of the theory will show, coordination is necessary among specialists and practitioners *in several different fields*. Parts of the following theory have been stressed by certain investigators, e.g., in Ferguson's (1956) use of "ability" and "learning set," Hebb's (1942) notion of A and B components in ability, Newland's[2] (1962) use of process and product, and Hayes' (1962) notion of "experience-producing-drives" (though the last joins cognitive and dynamic components here kept separate). But the necessary experimental conditions for constructive conclusions are possible only if the total theory is kept in focus, as is happily the case in, for example, the recent work of Horn (unpublished). The interrelated aspects of the theory may be briefly stated as follows:

1. That among the primary abilities in the area semantically designated "intelligence" in our culture (and which, as Guilford points out, is not *totally* general to cognitive performance) there is not *one* "general ability" second-order factor, as in the Spearman-Thurstone resolution (Cattell, 1943), but more. Two of these are highly cooperative (Cattell, 1952) in the sense that they agree in loading positively most of the general ability primaries and have largely zero loadings outside the intelligence field. Being cooperative they are very difficult to separate, and most previous ability factorings have been satisfied to recognize the joint factor as *one* factor. This conglomerate has been made the basis of IQ measurement and has been called "general intelligence." Technical reasons for this confusion are discussed in the section "Implications for Experimental Checks on the Theory."

2. The two factors, which will be called *fluid* (f) and *crystallized* (c) general ability, g_f and g_c, and indexed[3] as U.I. (T) 1 and U.I. (T) 2, have different properties, as follows:

a. Crystallized ability loads more highly those cognitive performances in which skilled judgment habits have

[2] In communicating to nonpsychologist teachers, for example, the particular aspect of the fluid and crystallized ability theory expressed in Newland's "process" and "product" proves to be readily conveyed by these terms. In any more precise and comprehensive context the terms mislead, however, for the *capacity to learn* is not only a function of the process (fluid ability) but often of the product, as well as of personality and motivation. Particularly when aids have been acquired, the rate of learning is a function not only of fluid ability but of the crystallized products of former application of process.

[3] In the Universal Index (U.I.) system proposed elsewhere (Cattell, 1957c; Cattell & Warburton, 1963), to preserve factor identification during changing explanatory development, U.I. (T) 1 was assigned to general ability and U.I. (T) 2 to flexibility of closure, in French's (1951) series. However, as pointed out elsewhere (Cattell, 1957a, p. 830), there is increasing evidence that flexibility of closure is really the Personality Factor U.I. (T) 19, outcropping in ability when ability tests only are used. Accordingly it preserves the historical importance of *order* in the index much better if the U.I. (T) 2 niche—flexibility going to U.I. (T) 19—is now allocated to crystallized general ability, and flexibility removed to U.I. 19.

become crystallized (whence its name) as the result of earlier learning application of some prior, more fundamental general ability to these fields. Thurstone's Verbal and Numerical primaries, or achievement in geography or history, would be examples of such products. Fluid general ability, on the other hand, shows more in tests requiring adaptation to new situations, where crystallized skills are of no particular advantage.

b. Before biological maturity (15–20 years of age) individual differences in the difference of g_f and g_c will reflect mainly differences in cultural opportunity and interest. Among adults these discrepancies will reflect also differences in age, since the gap between g_c and g_f will tend to increase with experience and the time decay of g_f. Although the speed of an intellectual performance is not a function of U.I. 1 and U.I. 2 alone, (for example, U.I. 22, Cortertia, is a powerful personality contributor; Cattell, 1957a), performances at speed, in any cognitive field, will load g_f more than g_c, while tests *without* time limit (unless they are absolutely new and adaptive), such as Thorndike called *power* tests, will have relatively large loadings of g_c.

c. Fluid and crystallized abilities will differ on their age trend plots for the general population, g_f reaching an early maximum at 14–15 years, while g_c increases to 18, to 28, or beyond, depending on the cultural learning period for the given subculture. Thereafter, whereas g_f will decline from about age 22 continuously to old age, g_c will show a later and lesser drop. In all these comments on g_c, it must be understood that we are talking about the *general* factor in a broad array of crystallized abilities, not any single crystallized ability.

d. The standard deviation of the IQ calculated "classically" from mental ages, will be about 12–16 points for g_c and 24–25 points for g_f (Cattell, 1937, 1955, 1957b). The latter will be more constant and biologically determined, whereas the former will hinge on culture habits. For example, classroom systems which slow the bright and press the dull will reduce the possible absolute individual difference in g_c at any age level, and therefore, g_c standard deviation of IQ, when annual increment is constant. Culture may similarly influence the sigma on g_c in that a powerful education, with larger annual mean increments and, therefore, larger age performance sigma, will tend to reduce IQ sigma. Parenthetically, since traditional intelligence tests are a mixture of g_c and g_f, the standard deviation of the IQ on such tests will be a function as follows:

$$\sigma_{IQ_c} = k\sqrt{\frac{\sigma^2_{f(p)} + \sigma^2_{c(p)} + 2r_{cf(p)}\sigma_{c(p)}\sigma_{f(p)}}{\sigma^2_{f(a)} + \sigma^2_{c(a)} + 2r_{cf(a)}\sigma_{c(a)}\sigma_{f(a)}}} \quad [1]$$

where p subscripts mean across people at one age, and a subscripts mean across age (years), taken as the variance of the means when 1 year samples are included across an agreed number of yearly age groups. As $\sigma_{f(p)}$ and $\sigma_{f(a)}$ can be assumed to be biologically given, the IQ sigma on a traditional, culture-contaminated intelligence test will in general become smaller (over school years only) where schools produce a large age increment in scholarship and where opportunity is not highly adapted to native ability differences, i.e., when $r_{cf(p)}$ is small.

e. For any same-age group the nature-nurture variance ratio will be much higher for g_f than g_c on the hypothesis that g_f is directly physio-

logically determined whereas g_c is a product of environmentally varying, experientially determined investments of g_f. At present the only available, and still scant evidence, reached with the new multiple abstract variance analysis method (Cattell, Stice, & Kristy, 1957) for culture-fair tests (which we shall show are largely g_f measures), does point to higher inheritance[4] for g_f. However, although it is our hypothesis that g_f is biologically and physiologically determined, as a function of total cortical cell count, this does *not* mean that one would expect anything like complete hereditary determination. For environment includes gestation period influences and later physical trauma and physiological change, all affecting g_f.

f. The reversible, day-to-day or month-to-month fluctuations in level will be different in g_f and g_c, the former varying only with general physiological efficiency (Pribram, 1960) and the latter less with physiology and more with recent exercise and interest.

g. The effect of general brain damage (e.g., arteriosclerotic) will be similar in both but more pronounced in g_f. The effect of *local* brain damage will be always to lower g_f in proportion to the magnitude of the damage, but the effect on the g_c score will only be through the *localized* ability changes, e.g., verbal aphasia (Lansdell, 1962). Consequently, if the g_c level happens to be estimated from a set of primaries which do not involve the particular area, as a representative, in the standard array of sampled areas in a traditional intelligence test, no significant fall in g_c IQ will necessarily be detected following local injury.

h. The implication in Part a above, that crystallized ability will be a general factor, i.e., covering a substantial total and variety of ability measures, needs to be examined. For such a factor to arise it is necessary that people (*a*) compete in, i.e., be interested and active over, the same series of experiential areas; and (*b*) experience difference degrees of learning opportunity (to acquire skills) *simultaneously* over all these areas. Evidence for the capacity of common instruction and experience to generate such group or general factors comes from the pioneer researches of Anastasi (1936) and of Woodrow (1938); and later work by Greene (1953), Fleishman (1954), and others. Only when the common variance of experience is considerable would we expect the magnitude (mean variance contribution over a standard sample of variables) of g_c factor to exceed that of g_f.

Since the present g_f levels of a set of people are simple growth-plus-accident functions of their g_f levels obtaining a few years earlier, and since their present g_c levels are only a somewhat more complex function of g_f levels over preceding years (see Equation 3), one would theoretically expect g_f and g_c to be substantially

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[4] This conclusion requires some special technical discussion for which we have little space Previous studies, such as those of Blewett (1954b) and Burks (1928), Freeman, Eysenck, and others used *twin* methods in which the environmental difference range (that between twins) is smaller than in the total family of sibs, as in our studies. Since the actual ratios we obtained with Culture Fair Tests are close to the mean of those they obtained with traditional tests, despite our environmental range being larger, the Culture Fair Tests can be concluded to have lower environmental determination. Our ratios were $\frac{122.5}{175.4} = 70\%$ genetic contribution for twins, and $\frac{83.3}{90.0} = 90\%$ for the genetic component in *between-family* variance in g_f (Cattell et al., 1957, p. 154).

correlated. The conditions favorable to most substantial correlation prevail in subjects at school age with respect to g_c factor measured over school subjects. In adult life, though school subjects will continue to correlate by "remnant correlations" (see below), a more valid measure of g_c will be made by weighting a much wider range of performances, though any one is likely to be less loaded than were school subjects. The use of culture-fair tests, to measure g_f, becomes, under these adult conditions especially, the more reliable, valid, and intelligible intelligence measure.

i. The rate of learning in any particular area will be a function, as far as cognitive components only are concerned, not only of g_f, but also of g_c, and the level of the specific ability factor in that area. In so far as g_c and the specific are concerned, we are thus saying that learning begets learning capacity. Even so, the effect of g_c as a general factor should prove to be small relative to g_f in proportion as the learning is in a completely new area. And in old areas the non-g_f-predicted variance is more likely to depend upon crystallized group factors than the general factor, namely, those associated with what we have called aids. The demonstration of such action in learning increments, however, involves complex and refined statistical treatment, as discussed by Humphreys (1959), Lord (1956), Tucker (1959), Woodrow (1939), and others.

IMPLICATIONS FOR EXPERIMENTAL CHECKS ON THE THEORY

The original theory of fluid and crystallized general ability has just been briefly recapitulated in the form of nine propositions integrating quite diverse fields of psychological observation, and indicating a possible array of tangible experimental research checks. However, there are further implications which cannot be stated concentratedly in the nine propositions and which require freer discussion, before the need for certain features in our own experimental design can be clear.

A first major implication is that the form of fluid ability—as a pattern of expressions and loadings—is due to an influence present and operative at the time of the experiment, whereas crystallized ability has a form determined by, and representing, history. Further, one is biologically, the other culturally determined. Our earlier analogy for g_c (Cattell, 1943) of the shape of a dead coral formation, representing the properties and accidental history of once living coral organisms, and Newland's (1962) recent use of "process" and "product" (or Guttman's "analysis" and "achievement") for g_f and g_c, bring out this meaning figuratively, in simple metaphors, but not quite as precisely or completely as the real scientific model needs to do.

In behavioral terms, the theory of g_f is that it is "a capacity to perceive relations and educe correlates" in Spearman's (1932) original sense. Further the theory postulates that relations, beginning with those between sensory elements, can be arranged objectively in a hierarchy[5] of increasing abstraction and

[5] The present writer's definition of intelligence proceeds to developments of Spearman's relational theory which should be distinguished from those for which Spearman would be responsible. It assumes that the hierarchical organization of relationships among relationships will begin with sensory fundamentals. The special sense areas will therefore form subsets, in the sense of mathematical set theory, and in a psychological sense, in larger sets. From this one might be inclined to conclude that the most complex relations must necessarily lie in the Cartesian space involving all six sensory sets. This *may* be, but at least for the ordinary ranges of intelligence, sufficient complexity can be reached in the hier-

complexity, and that the level of g_f of an individual at a given time is defined by the position in the hierarchy at which his insight fails, as shown by the onset of chance response behavior. It is a task for logicians (Braine, 1959), ecologists, and machine-simulation specialists to develop bases for this hierarchy; but our hypothesis would be that it will be found to rest on sensory area fundaments, and therefore, require for its genesis intact sensory neurological structure, as stated in Footnote 5. Meanwhile psychologists can delineate a rough hierarchy by trial-and-error item construction, ordered by enlightened difficulty-level analysis (extremity-vector analysis; Cattell, 1957, p. 360) experiments.

Most attempts, operationally, to measure intelligence, even when it has been conceived as some sort of constitutional, fluid-ability, relation-eduction capacity, have been made, however, through first measuring crystallized ability (i.e., on fundaments and even relations upon which strong cultural training has already occurred). The new movement which began with studying relation-education in perceptual tests (Cattell, 1931, 1940; El Koussy, 1938; Fortes, 1932; Line, 1931; Penrose & Raven, 1936) apparently supports the theory, however, that quite complex intelligence-demanding relations can be successfully set up in entirely new (and therefore *perceptual*, not memory-stored) material. Both experience and psychological reasoning in this area of endeavor lead to the general conclusion, however, that to place individuals in their proper rank order on g_f it is necessary that the test fundament material should be either, (*a*) equally overlearnt by all, or (*b*) equally new to all (and therefore "perceptual" in the sense used in this particular field). Anything in between will contaminate the variance with experiential variance. The sensory area, e.g., whether an analogies test is visual or by sound, seems unimportant. However, the approach through *a*—equally overlearnt—is risky unless certain conditions are watched, and even though it *ranks* persons correctly, it gives no correct account of absolute level. Moreover, there is always the risk that individuals highly familiar with and trained in a field will know *by rote* the correct answers to relational decisions into which they do not have, and have never had, any insight. For example, many people can respond that Einstein's famous formula is $e = mc^2$, not $e = mc^8$.

With these introductions to the general theoretical position we are ready to bring it to precise formulation, and to ask what the implications are for a crucial factor analytic experiment. At this level of precision it can no longer be assumed, at least in the ordinary circumstances of experimental testing of an intellectual performance, that performance level is going to be determined *only* by fluid and crystallized general ability. The specification equation must put these hypothesized two general abilities in the full context of specific abilities and temperamental and dynamic traits, as follows:

$$p_j = s_{j1}F_1 + s_{j2}F_2 + s_{jn}F_N + s_{jt}T_T + s_{si}T_I + s_jF_J \quad [2]$$

where F_1 and F_2 are g_f and g_c, respectively (as in the U.I. factor index); the Subscripts T, I, and J represent

archy within the relations arising from *one* sensory area, to reach the limen of failure of insight. This happens because there is a very large number of further subsets among which complex relations can hold, even within one sensory area. The successful use of Culture Fair Tests (involving largely spatial *fundaments*) but transcending merely spatial relations and ability, offers some proof of a sufficiently high relational complexity being attainable upon a limited (two sense) sensory foundation.

temperament, interest (dynamic, motivation), and local area (specific) ability factors, respectively, and F_N represents a factor of strength of immediate memory (for if, as in senility, the capacity to hold the material is deficient the analytical capacities cannot act upon it).

Actually these latter, in a fully expanded formula, would each be several factors, but that expansion is unnecessary for our present statement of essentials. If, moreover, the performance (p_j) is made a highly intellectual one, and there is good, uniform, experimental motivation, and sheer speed is not so important that it brings in temperament factors, the analysis of the variance in p_j can be restricted largely to the three cognitive factors.

To understand better what would be the theoretically expected outcome of a factor analysis of purely cognitive variables (i.e., of "intellectual" performance measured under such conditions) let us, however, pry more closely into the nature of F_2. This is our specification equation notation for g_c which, according to Propositions e and i, should itself be resolvable *historically* into other factors. In fact, any variable likely to be highly loaded on g_c might, by special factor analysis directed to splitting F_2, be expected to yield factors consistent with the Equation 3. This equation states that a crystallized ability performance (p_{jc}) is a function of previous time (T) and interest (I) invested in the application of fluid ability; memory; and specific, problem solving aids.

$$p_{jc} = w_{jT}(T_e + T_{ej})w_{jI}(I_e + I_{ej})$$
$$[s_{jl}F_{le} + s_{jM}F_{Me} + s_{jA}F_{Ae}] + C_j \quad [3]$$

where:

- T_e represents the time given earlier (e) to applying one's abilities in a *whole* field of intellectual performances—i.e., to p_{jc}, p_k, and the numerous other performances covered by F_2;
- T_{ej} is time specifically given only to p_{jc};
- I_e represents the mean strength of interest in learning at that earlier time which, like T, is assumed *common* to several p_{jc}'s—e.g., those in school work;
- I_{ej} is interest specific to p_{jc} only;
- F_{le} is the level of fluid general ability over this *earlier* (not the present) learning period;
- F_{Me} is a memory capacity factor, also at its earlier level at that time;
- F_{Ae} is an aid, i.e., a discovered "instrument for problem solving" mentioned earlier[6] which operates as a group factor over some area less wide than the general ability factors;
- C_j is a statement of the change, an increment or deterioration by

[6] An aid is a term for the generalizable habit, discussed by Piaget (1947), Hunt (1961), Hayes (1962), Forgus (1955), Ferguson (1956), and others which opens up faster learning in a whole region, e.g., the infant's grasping the possibilities of babble, the kindergartener's discovery that doing to others as one would like to have done to oneself is an avenue to social goodwill, the junior high school student's realization that a triangle drawn on the board means not that particular triangle but a class of triangles, or that letters in algebra can stand for numbers, and so on. Such a piece of mental equipment stands in the class of achievement-attainment measures, as the use of F_A is intended to remind us. In the last resort, it is itself a product of F_l, F_M, etc., and belongs to that subclass of attainments which have the special property of aiding further attainment. As such it has an independence of these other terms and a characteristic field of operation, probably as a group factor like a Thurstone Primary Ability. In view of the length of time which Piaget and his followers have been discussing such action it is surprising that there are still no multivariate experimental studies demonstrating their existence, and their nature and position among abilities generally; but they seem a reasonable speculation.

"forgetting" since the original learning experience, due to time and changing interests, retroactive inhibitions, etc. (In the scholarship area one may think of the effect upon a foreign language area of neglect for a decade.)

The s's are the usual situational indices showing by weights specific to the time, interest, ability, etc., how much intelligence, memory, etc., contribute to performance, p_{jc}. The w's are weights similarly given to number of repetitions (time of exercise) and strength of interest, e.g., w_{jI} is the rate at which increased time expenditure brings increase in p_{jc} score.

Two statements in this formula need especial comment: (*a*) Whereas in a present performance the specification equation for the factor analytic model *adds* factors of ability and interest to estimate the performance, we have here multiplied ability, memory, etc., by interest and by the time, recognizing thereby a different quality in them from additive influences within the organism. Neither the rather hollow science of learning theory (still helpless where human higher abilities are concerned) nor the degree of precision of factor analysis (Tucker, 1959), permit decision yet as to whether an additive or product model is better. But the product formulation is consistent with our formulations of motivation in the *dynamic calculus,* elsewhere (Cattell, 1957a) and is contingently adopted for these *accumulative learning effects.* (*b*) The T and I terms as indicated by their appearance outside the bracket, are not given separate values for the g_c, memory factor and aid factors, though in an expanded formulation they might be given different weights. By the nature of the learning situation intelligence, memory, etc., would normally operate together so they are gathered within the brackets with T and I outside. (The TI, if one wishes, is an integration of interest over time.) The introduction of the memory term (F_M) recognizes, as pointed out earlier, that a subject's correct judgment, even on a test item of a highly intellectual kind, may depend on goodness of rote memory, operating over the course of exposure to an intellectual education. For example, a correct decision in geometry may depend on remembering the formula for the area of a circle, or solving a crossword puzzle on good rote memory for Greek. As Cohen (1957) shows, the variance in intelligence performances from this memory component becomes large in older adults.

Now our problem is to ask to what kind of a factor structure the theoretical, psychological formulation in Equation 3 would be expected to lead. Should we expect distinct factors for T and I? The answer seems to be that although very specially planned and exceptionally executed factor analyses might express a product relation like $T_eI_eF_{Ie}$ as an addition of three factors, the normal precision and choice of variables would yield a single general factor loading most highly the variables which have simultaneously shared high interest, much time and much involvement of fluid ability. However, because they are multiplied by the same T_eI_e the intelligence, memory, and aid factors would tend to be cooperative (Cattell, 1952) (i.e., loading the same variables). And if the variance in T_eI_e were great, we should expect a powerful second-order factor among them defined by T_eI_e. Thus there are really two possible definitions of the crystallized general ability factor: (*a*) As the first-order factor $T_eI_eF_{Ie}$, representing the past areas of investment of fluid ability, and (*b*) as the second-order factor—strictly a general achievement factor—corresponding to the in-

vestment of T_eI_e in fluid ability expression, memory, and various aids and special primary abilities.

The situation is further complicated by the fact that skills not only increase but decay, and a common history of change or decay *after* the pattern has been reached as in the first part of Equation 3 could also lead to a common factor structure. This, with some over-condensation, we have represented by the term C_j in the latter part of Equation 3. For example, if C_j is a time function and our sample is scattered from 30 to 60 years of age, a general factor would tend to appear over all school subjects due to their all being forgotten to the same degree, as a function of the time lapse. Such considerations have been grossly overlooked in traditional adult intelligence tests, as well as the fact that a new general crystallized ability factor, distinct from the scholastic one, is likely to arise through common learning by the application T_eI_e to whatever common areas of g_f use exist for adults in our culture. Here only culture-fair, but not traditional, intelligence tests seem to recognize that in the majority of our population which proceeds to "nonverbal" occupations (e.g., garage mechanics, farmers, etc.) a substantial decay of the attained crystallized ability levels in vocabulary and similar areas supervenes (Burt, 1955). Further, the normal adult intense investment of skills in relatively narrow occupations will tend to make the appearance of a second "adult" crystallized ability factor, superseding the school curriculum crystallized general ability factor, a relatively poor substitute, of lower variance and poorer predictive power.

In summary, one must face the complication that the crystallized general ability factor is for various reasons Protean. Strictly the term crystallized intelligence (Symbol g_c) should be restricted to the first of the two factors mentioned three paragraphs above, namely, to the Expression $T_eI_eF_{Ie}$. The second-order factor covering $T_eI_e[F_{Ie}+F_{Me}+F_{Ae}]$ is in any sense of the word a *general achievement* factor, though it must be confessed that a substantial part of traditional intelligence test scores derive from this source. Both of these factors are at the mercy of culture, and even before one encounters the difficulty of cross-cultural comparisons thereon, one gets entangled in the shift which occurs in them between the schooling age and the adult activity age. The Protean character is thus at best reducible to four concepts, corresponding to four crystallized general ability factor patterns.

1. School age crystallized intelligence (g_{cs})—common to g_f expressions powered by time and interest within the culturally set framework of the school curriculum.

2. School age crystallized achievement (a_{cs})—a second-order factor across intelligence, memory, special ability areas, etc., corresponding purely to the effect of common time, interest, memory, and curriculum. It will differ markedly from g_{cs} in being predictive of achievements (e.g., athletics, foreign languages, and rote areas of school learning) in which g_f has never had any appreciable importance or role.

3. Adult activity crystallized intelligence (g_{ca})—representing whatever skills develop strictly from fluid intelligence being applied in whatever activity areas receive common time and interest from adults, i.e., $T_aI_aF_{Ia}$.

4. Adult activity crystallized achievement (a_{ca})—representing whatever general factor is produced in adult achievements through their common expenditures of time and interest. Except in mixed samples from different cultures

this would be a very tenuous factor of little practical predictive use.

Only in the case of Concept 1 would we expect the correlation of fluid and crystallized intelligence factors to be really substantial, and the term intelligence is best eschewed completely for Concepts 2 and 4. Also the terms "school practised" and "non-school-practised" abilities are misleading for g_c and g_f, for the first would describe both Concepts 1 and 2 and the second could be either Concepts 3 or 4, or g_f. Any combination from decay influences is omitted from this reduction to four general factor concepts.

DESIGN OF THE EXPERIMENT IN RELATION TO EXPECTATIONS

Testing the theory of fluid and crystallized intelligence requires not one experiment but a comprehensive organization of experiments. Yet one which needs to be performed before all others, as crucial to further work, is a factor analysis to check whether one factor or two runs through most intellectual performances.

Technically, two questions need care in planning such a factor analytic experiment:

1. Are we going to aim at abstracting $T_eI_eF_{Ie}$ as a single factor or at splitting it? Theoretically it is not beyond the power of fine factor analysis to split a presently operating whole into the components which historically produced it. A product relation would be approximated by additive factors as the plasmode studied by Bargmann (1955), Cattell and Dickman (1962), Cattell and Sullivan (1962), and Thurstone (1947) sufficiently demonstrate. But in the present case T_e, I_e, and F_{Ie} would be cooperative, because by the necessary choice of tests for intelligence they would be variables also on which common T_e and I_e have been expended. Besides there is much to suggest that product relations are often revealed by cooperative factor patterns. Consequently, since cooperative factors are hard to rotate correctly, it would seem best at this stage not to attempt such finesse but to accept variables sufficient to reveal crystallized intelligence as a "factor in being," if it so exists. When such deeper analysis is eventually undertaken it would be well to keep in mind that I_e might split again into a number of personality and dynamic factors (e.g., super ego strength, emotional stability) affecting the investment of fluid intelligence in crystallized intelligence skills. Essentially this has recently been powerfully argued by Hayes (1962). Decay factors will also need consideration.[7]

2. Provisions must be made for definition by rotation of *factors possibly general to every cognitive test employed.* One reason why the present theory, rather than the Spearman-Thurstone-Thomson theory of a single general intelligence factor, has not hitherto received due and serious consideration is that most workers in the cognitive field have included only cognitive tests and thus rendered quite impossible any revelation of a distinctly rotatable second

[7] A summary of the main influences now posited in any performance in the area semantically designated intelligence might be convenient at this final step: (*a*) a fluid *general* intelligence factor; (*b*) a crystallized general intelligence factor; (*c*) one or more crystallized auxiliaries, corresponding to time and application acting with sheer capacities to memorize at the former among time (e.g., knowledge of vocabulary as distinct from nice judgment of meaning among familiar basic English words); (*d*) factors corresponding to any general patterns of change or decay since learning; (*e*) existing personality and dynamic (motivation) factors helping application to an intellectual test.

Here *b*, *c*, and *d* have structure in virtue of history, while *a* and to some extent *e* are structures due to present organismic functional unities.

general factor. The procedures of Burt (1955) and Vernon (1950) for example, neither permit unique determination of the alleged general intelligence entity nor separation of a possible second general factor.

The technical failure in factor analysis here, by which investigators have failed to see the wood for the trees, is only a special case of ignoring a general principle in scientific method—namely, the figure-ground principle, according to which, if we wish to define X adequately we must introduce data which is not X. In this case *to rotate with any approach to precision a factor expected to be general to all abilities one must include a wide framework of nonability factors (not merely variables) as hyperplane stuff* (Cattell, 1952). Indeed, in spite of the immense importance of the intelligence concept to education, the second-order general ability factors appear never to have been so rotated, in any research discoverable by the writer, to a comprehensively determined simple structure position.[8]

In the present study we planned to reach the second-order analysis with only nine ability measures—a bare but sufficient minimum for testing the main cognitive hypotheses—but with 13 nonability measures. The latter were chosen (*a*) to outnumber abilities, (*b*) to be well-known *factors*, not merely variables, so that they would fan out into a maximum number of dimensions to give stability of rotation of the ability factors with respect to many reference points, and (*c*) themselves to have a known second-order structure, the emergence of which in the present blind rotation would give a check on its general correctness. These requirements[9] we felt to be best met by taking Personality Factors A through Q_4 (having the same meaning as on the 16 PF Test) on the High School Personality Questionnaire (HSPQ; Cattell & Beloff, 1959).

The nine ability variables consisted of five Thurstone Primary Abilities—Verbal, Spatial, Reasoning, Number, and Fluency—and four subtests from the Institute of Personality and Ability Testing (IPAT) Culture Fair Intelligence Test, Scale 2a—namely, Perceptual Series, Perceptual Classification, Matrices, and Topology. If our hypothesis is correct all the Thurstone primaries, with the exception of Spatial and possibly Fluency preformances should load on a crystallized ability factor (as also should be the case with each specific aid factor).[10] On the other hand, the Culture

[8] In many cases investigators have shown a curious inconsistency in believing that first-order factors should be rotated but second orders should be left where they fall! (The Schmidt-Leiman determination of loadings of variables on second orders, for example, assumes this.) In other cases, the explanation seems to be the usual exhaustion phenomenon, in which, by the time the investigator gets to the second order of a long factor analysis, neither time nor energy remains to press on to a determinate second-order resolution. Mainly, however, it seems that like early climbers on Everest, they have not dragged along at the early stages of the expedition the equipment which foresight might have indicated to be essential at the last stage of ascent.

[9] This procedure of entering the ability research with more personality than ability measures, may seem quixotic to conservative researchers, but in addition to the reasons given above it is actually strategically necessary in terms of demonstrating that second-order ability factors are in fact *distinct* from personality factors. This can by no means be taken for granted, since some "primary abilities," e.g., flexibility of closure, now appear (Cattell, 1957a) as outcroppings of personality factors "in disguise" in the cognitive realm.

[10] A small note on a large issue is necessary here in connection with the notion raised above that an accumulation of separate aid developments might constitute a general factor. The magnitude of the aid factor, $F_A{}^1$ is a joint function of F_1 and some environmental "accident" A^1, though whether it should be written as (*a*) $F_A = F_1 \times A$, or, (*b*) $F_A = F_1 + A$ is uncertain. In either case A would be operating as

Fair subtests, being purely perceptual and lacking even pictorial reference to the not-present should load only a fluid general ability factor. However, since even the Verbal item decisions involve some immediate adaptive performance, the fluid factor should extend to some degree (in the rotation position defined by the wider realm of personality measures) into such performances as Verbal, Spatial, etc., primaries.

The choice of sample was made on the assumption that Horn's (unpublished) research would use adults. For a complete answer on the above hypothesis requires a check on the expectation that the crystallized intelligence factor will behave very differently (as g_{cs} and g_{ca}) with children and adults. There is some advantage in beginning with, say, 12-year-old children, because we may expect less complication from any decay term (C_j in Equation 3). Also one might expect the g_{cs} factor, based on so uniform an area of earlier application, to be larger than with adults (about the same variance as the g_f) and thus easier to locate in a first study.

DATA AND ITS ANALYSIS

The sample consisted of 124 eighth grade boys and girls from Paxton, Illinois, and 154 from the seventh and eighth grades of a junior high school in Springfield, Illinois, a total of 277 (i.e., one dropped) being carried forward to the analysis. The Thurstone Primaries (except for Fluency) were given in both forms, and HSPQ was similarly given in separate A and B scores for each primary personality factor; but the Culture Fair Intelligence Test used only one form (which must be remembered in appraising later saturations). Testing took about 3 hours and was under one examiner.

The *first* order analysis required product-moment correlations[11] for 44

a group factor covering from two or three to a large number of performances.

In the early stages of the operation of an aid, as suggested above, it should be detectable in either case as a group factor. In later stages, it would be possible for an accumulation of overlapping group factors to be resolved as a general factor, indeed, *they would constitute part of the crystallized general ability*. That a set of overlapping small group factors could alternatively appear as a general factor has puzzled factor analysts since the early debates of Thomson and Spearman on the interpretation of g as a "sampling of bonds" versus a "single power." This factor analytic problem can be kept clear only by invoking the *principle of organismic setting*.

This principle states that although each of a set of separate covariational entities may appear as a separate group factor, yet, if they all have some quality in common, so that they can appear as part of a larger organization, then, by simple structure resolution, their *sum* will also appear as a single factor, in the context of several other, qualitatively different, such accumulations. Thus although each leaf in a heap of leaves is a separate entity and (if its variation, e.g., in perception from various angles, were considered, a separate factor) yet the heap also has its dimensions, as an entity, among other heaps. Similarly for cells in an organism or people in a team. What one gets factor analytically depends upon the organismic level at which measurements and rotations are made.

Even before one gets a shift of organismic level, a mere change of scale, from microscopic, or, in terms of psychological variables, from low to high density of representation of variables, can convert the resolution from a multiple group factor outcome to a general factor outcome. Pending further experiment our hypothesis is that the variance produced by cummulative action of distinct aids will, except for what is left in the truncated group factors (Cattell, 1962) become part of the crystallized general ability factor, i.e., F_1 and F_4 in Equation 3 will both go into its variance. This states an important difference from Hunt's (1961) theory.

[11] The correlation matrix, centroid, lambda transformation, and final simple structure matrix have been deposited with the American Documentation Institute. Order Document No. 7381 from ADI Auxiliary Publications Project, Photoduplication Service, Library of Congress; Washington 25, D. C., remitting in advance $1.25 for microfilm or $1.25 for photocopies. Make checks payable to: Chief, Photoduplication Service, Library of Congress.

variables (9 Thurstone abilities, 4 Culture Fair subtests, and 28 HSPQ personality variables), i.e., in all 41 plus 3 random variables for hyperplane testing. Since they consisted mainly of paired markers for factors, it was not surprising that the Tucker (Thurstone, 1938) and Guttman (Cattell & Sullivan, 1962) tests (separately applied) pointed to 20–24 factors to be extracted. We settled on 22 as the limit for determinate, converged, iterated communalities and iterated four times to reasonable stability.

The rotation for simple structure was carried out with extreme care, since as will now become apparent, all conclusions in the second order hinge upon the angle values settled upon in the first. Simple structure reached, after four overall rotations, the very high value of 82% of all variables lying within the $\pm.10$ hyperplane. At this point in the blind rotations an inspection showed that the usually accepted structure for both the Thurstone Primary Abilities and HSPQ primary personality factors had emerged, independently, with excellent clarity. The two prescribed markers, and virtually nothing else, loaded each named factor. Although only one marker had been inserted for Fluency and each Culture Fair subtest, these five picked up enough on other variables, e.g., classification had loading on HSPQ Factor B, for each to appear as a nonspecific factor, though they are in the usual wider sense, specifics.

The C_R matrix for this carefully rotated study was now inverted to give the correlations among the 22 factors ($C_F = DC_R^{-1}D$) and this was again factored. The advantage of proceeding to the second order from the C_R instead of from correlations of actual battery measures of the factors is that we eliminate distortions from the unreliabilities and imperfect validities of actual scales. In exchange we pick up whatever uncertainties remain in the simple structure rotation, but with at least 20 variables to define the hyperplane, this should be a lesser risk and offer a better design.

Centroid factoring of the first-order 22-factor matrix yielded, by the same pair of criteria as before, 8 factors. Their simple structure rotation is shown in Table 1, together with C_F of their correlations (see Footnote 11). The simple structure here is no longer quite as clean as at the first order. If the oddity of Factor 7 is omitted (with which we are not much concerned) the count is, however, still 59%, which is high among reported studies and at any rate the essential maximum or plateau attainable among present possible rotations.

The structure of Table 1 reveals unmistakeably the distinctness of the fluid general ability pattern, which is Factor 1 (loading the four subtests of the Culture Fair, the Thurstone Spatial, and the HSPQ Ego Strength, C, factor), and the crystallized general ability factor, Factor 2, which loads Verbal ability, Reasoning, and Number most highly. That these two ability factors, roughly orthogonal to the remaining six, are essentially in the right position is attested further by the correctness of the second-order factors emerging among these latter personality factors. Factor 3 is the typical U.I. Q. I. (Cattell, 1957a) second-order extraversion factor (Carrigan, 1960) covering the primaries A, D(−), F, H, Q_2(−). Factor 4 is the general Anxiety versus Adjustment factor—C(−), O, and Q_4(−), (with Q_3 slightly anomalous); and Factor 5 continues these previously recognized second-order factors to include that indexed as U.I. Q. III (Cattell, 1957a). Although these are to be remarked here only as confirmation of the "setting" for the two main ability factors, it is possible that Factor 6—"super-ego-good-upbringing" or Control—has also some interaction with crystallized ability.

Let us for the moment postpone ex-

TABLE 1

SECOND-ORDER FACTORS AND THEIR INTERCORRELATIONS

Part A: Factor loadings (in reference-vector correlation form)

First order factor variable	F_1 g_f	F_2 g_c	F_3 Exvia-Invia	F_4 Anxiety	F_5 U.I. Q. III	F_6 Control	F_7 ?	F_8 ?
Thurstone primaries								
Verbal	.15	.46	−.17	−.03	.04	.07	−.05	−12
Spatial	.32	.14	−.02	−.03	−.03	.05	−.27	.04
Reasoning	.08	.50	.08	.02	−.02	.02	.38	−.14
Number	.05	.59	−.04	.05	−.07	−.05	−.10	.03
Fluency	.07	.09	−.52	−.18	.12	.26	.13	−.07
IPAT Culture Fair subtests								
Series	.35	.43	.05	−.12	.06	.10	.24	.04
Classification	.63	−.02	−.05	−.05	.00	.14	.16	.06
Matrices	.50	.10	−.23	.04	−.01	−.08	−.25	.42
Topology	.51	.09	.16	.12	.03	−.03	−.14	−.00
IPAT HSPQ								
A Cyclothymia	−.04	.52	.31	−.23	.05	.11	.20	.03
C Ego Strength	.21	−.07	−.04	−.52	−.08	−.09	−.49	.12
D Excitability	−.04	−.44	−.44	−.09	.09	.04	.14	−.27
E Dominance	−.15	−.01	−.05	−.00	.61	−.04	−.02	−.17
F Surgency	−.05	.09	.39	−.07	.06	−.40	−.20	−.09
G Super Ego Strength	−.14	.08	−.01	−.10	−.10	.43	.41	.05
H Parmia	.21	−.04	.55	−.17	.00	.13	.03	−.03
I Premsia	−.09	−.29	−.05	.26	−.54	−.05	.43	−.01
J Coasthenia	−.10	−.04	.01	.39	−.12	.45	.01	.16
Q Timidity	.16	−.01	.01	.73	.09	−.04	.10	.17
Q_2 Self-Sufficient	−.06	.05	−.37	−.05	−.43	−.04	−.04	.05
Q_3 Self-Sentiment Control	.05	−.02	.06	.00	.32	.35	.56	.11
Q_4 High Ergic Tension	−.04	.37	.03	.45	−.00	−.37	−.14	.05
% in ±10 hyperplane	.51	.60	.60	.51	.73	.55	.32	.60

Part B: Factor correlations

1. Fluid general ability	1.00	.47	.29	.35	.09	.21	−.01	−.15
2. Crystallized general ability	.47	1.00	.17	.15	−.03	.28	.03	−.20
3. Extroversion versus Invia	.29	.17	1.00	.17	−.01	.10	−.24	−.05
4. Anxiety versus Adjustment	.35	.15	.17	1.00	.24	.06	.14	−.12
5. U.I. Q. III	.09	−.03	−.01	.24	1.00	−.46	.06	.26
6. Control	.21	.28	.10	.06	−.46	1.00	.07	−.70
7. ?	−.01	.03	−.24	.14	.06	.07	1.00	−.11
8. ?	−.15	−.20	−.05	−.12	.26	−.70	−.11	1.00

perimental examination of our assumptions about the inherent modes of expression of these two factors as shown by loadings and pause to examine their mutual relation as shown in Tables 1 and 2. The substantial positive correlation, +.47, shown in Table 1, Part B, is what would be expected from our theory, on the grounds that both must be highly related to an entity which existed a few years earlier, and which we have symbolized (Equation 3) by F_{1e} (or g_{fe})—the fluid ability level earlier. The present crystallized ability level—F_2 (or g_c)—is a function of F_{1e} and the time and interest applied at that time.

The present fluid ability level—F_2 (or g_f)—is an organic continuation of F_{1e}, falling short of perfect correlation with it only because different times have elapsed for different people and the interlude has treated them more or less kindly in terms of brain damage, etc.

If next we ask what this correlation implies in terms of a third-order analysis, different theories will give different answers. Our theory would posit that F_1 and F_2 (g_c and g_f) should come together in a single higher order factor, because historically a single influence, F_{1e}, is responsible for them both. It has, so to speak, fathered F_1 out of time (as "the child is father of the man") by simple maturation, and F_2 out of experience by interest and learning investment. Whether the third-order F_{1e} should load higher on F_1 or F_2 depends on many things. High similarity of experiences (including time length) and interest should raise the loading on F_2, whereas heterogeneity of the group in this respect should lower it. Heterogeneity of the group in age and time lapse since the formative period of g_c should lower the loading in F_1. In a group of school children such as the present, in the midst of their growth period, the F_{1e} loading in F_1 should be quite high and that in F_2 a little short thereof. In adults or children more heterogeneous for age, the loadings should both be lower.

The third-order analysis designed to test these hypotheses again had, by reason of the initial planning, enough hyperplane stuff to determine the rotation of this still broader general ability factor. Further, the rotation was again made with pure factor measures, not error-contaminated actual battery scores, for simple structure was attained in the reference vector matrix underlying C_F shown in Table 1. In this C_F four factors were indicated by the Tucker test and a satisfactory 63% in the hyperplane was obtained after five overall rotations, with results shown in Table 2.

TABLE 2

THIRD-ORDER FACTORS AND INTERCORRELATIONS

Second-order factors	Third-order primary factors 1.	2.	3.	4.
Fluid intelligence	.69	.02	—.07	.00
Crystallized intelligence	.63	—.04	.07	.32
Extraversion versus Invia (U I. Q. I)	.18	.03	.38	.23
Anxiety versus Adjustment (U.I. Q. II)	—.01	.01	00	.79
U.I. Q III	.09	—.51	—.07	.32
Control	.01	.99	—.03	—.05
F_7	.06	.03	—.74	.04
F_8	.02	—.69	08	—.06
% in ± 10 hyperplane				
Primary factor correlations at third order				
1.	1.00	.38	.14	.25
2.	.38	1.00	.00	.10
3.	.14	.00	1.00	—.16
4.	.25	.10	—.16	1.00

If our hypotheses is correct it would not be right to conclude from Table 2 that "fluid and crystallized general ability form a single third-order factor," but rather that a single influence, which is fluid ability as it stood during the formative period of crystallized ability, is causative to the present levels of both. That this "formative fluid ability" influence is in fact F_1 in Table 2 could be checked by preserved records of the subjects' earlier fluid ability scores. It is noteworthy in Table 2 that this fluid ability factor has no significant loading on any personality factor, but stands as a measure of pure ability. On the other hand, it has slight, significant correlations *as a factor* with control and anxiety of a kind we have come to expect from the linking of intelligence with other factors through the natural selection of social status (Cattell, 1945; Fisher, 1958). The factor contingently dubbed Control here has, on the other hand, significant relations to crystallized ability in Table 1 and must be considered one of the personality factors contributing, partly by aiding school memo-

rizing, to crystallized ability, via F_M, etc., in Equation 3 above.

NATURES OF FLUID AND CRYSTALLIZED INTELLIGENCE

With concepts defined and checked at the experimental level we can now proceed to broader formulations. The statements so far are:

1. That intelligence test and school performances must be considered, more comprehensively than hitherto, to be determined, at the moment, by personality and motivation factors as well as by two distinct second-order general factors, (fluid and crystallized intelligence) as symbolized in Equation 2. Equation 2 is primarily an expression for performance in a timed test, but with trivial adjustments, is also an expression for rate of learning.

2. As a result of this mechanism of learning operating uniformly across school subjects, and other areas, for time intervals differing for different people, third-order general factors of school achievement (for children) and adult activity achievement (for adults) are created. (There is no space to discuss their relation here, but they should be slightly positively correlated.) The term crystallized intelligence, however, is semantically better reserved for the second-order factor in such performances as derive their positive correlations only from the action of fluid ability, not from memory and various special abilities operating in the school (or adult) settings. Crystallized intelligence and achievement are thus *not* identical general factors, the former being narrowed in manifestation to the handling of complex relations, whereas the latter could cover whatever the breadth of the cultural schooling curriculum embraces.

A special logical complexity arises here, however, in that rate of learning in complex fields is assisted both by fluid and crystallized general abilities, as well as by such aids (usually primary, group factor patterns) as are relevant to the particular field (Equation 3). Consequently, crystallized ability must be considered to be begotten by crystallized ability as well as by fluid ability. This may seem to contradict the concept above that crystallized ability is a consequence and function of fluid ability levels, but it is not so if we consider (*a*) that fluid ability is so to speak put out at compound interest, and recognize also that the rate of return may be different over different periods of learning, depending on interest, etc.; and (*b*) that crystallized ability is not *only* a function of fluid ability but also of personality factors, etc. Thus personality factor "deposits" occur in what at the time of learning is considered an ability —crystallized general ability—and also in aids. If space permitted, these relations of earlier personality and motivation factors, as operating in Equation 2 to crystallized abilities as in Equation 3, could be worked out in a fourth or "historical depth" equation.

As to the descriptive similarities and differences of g_f and g_c, both are very general to complex, intellectual, relation-perceiving performances, and even in regard to the variables chosen to discriminate them one notices, in Figure 1, that each has some tendency to load the variables set to mark the other. However, the slight loading of crystallized intelligence in say Fluency is not as great as that of fluid ability in Verbal or Series. (Note with oblique axes the lines of projection are drawn parallel to the axes in Figure 1.) Parenthetically, it will be noted that if the experiment had been undertaken without personality factors, etc., as hyperplane substance, the rotator would almost certainly have fallen into the false resolution of making these factors more highly correlated

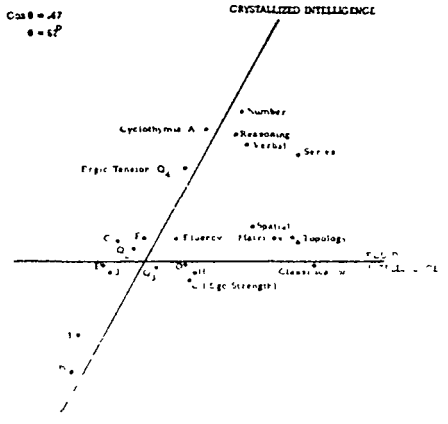


FIG. 1. Culture-fair subtests and Thurstone Primaries in relation to g_f and g_c. (Rotation assisted by 16 PF factors in hyperplane.)

(especially in the absence of D, I, A, and Q_2) by going through the ability variables only, e.g., through Verbal and Topology.

Figure 1 also confirms the whole argument of Equation 2 to the effect that crystallized intelligence, being partly the product of motivational and personality history, will have more significant association with personality factors. This association is not only statistically significant but makes psychological sense, for crystallized ability correlates substantially positively with Cyclothymia (Factor A in HSPQ), and negatively with Excitability (D), and Premsia (I) which other researches show to be similarly related to general school achievement. Similarly in the motivation realm (Connor, 1961), docility (low self-assertion) and super ego strength have the highest correlation with total school achievement. Their correlations with the Verbal (V) factor, the Reasoning (R) factor, and the Culture Fair are, respectively, .20, .48, and .02; and .19, .41, and −.02, (see Footnote 10) the Culture Fair (fluid ability) measures consistently being more free of association with personality factors affecting achievement. As stated above, this greater freedom of the second-order fluid ability from personality loadings, which supports our hypothesis, is not incompatible with its having correlations therewith at the third order, pointing to genetic and social status ties.[12]

A word is necessary here to forestall a misunderstanding which will almost certainly arise in the attempt of conservatives to avoid any major restructuring of ideas. This will be the view that g_f is nothing more than the k or "practical ability" factor, located by such factorizations as those of Vernon (1950) and Cohen (1957). There are three good reasons for rejecting this. The first, comparatively trivial, is that k has always shown decided associations with masculinity and mechanical experience whereas in our mixed boy and girl sample g_c shows statistically no difference and is actually a shade higher for the girls. Secondly, and more systematically, the whole mode of factor resolution in the Burt-Holzinger-Vernon bipolar system is quite different from multiple-factor simple-structure resolution and never yields a one-to-one matching. The former method puts verbal ability on one side and practical-mechanical on the other. In the present writer's opinion this is in any case a fallacious form of factor resolution, for since the first general factor is indeterminate, the whole hierarchy is fictitious. But, fallacious or not, it is, except by some accidental

[12] In looking for indications of some separate action of the interest-personality factors posited in Equations 1, 2, etc., one may note in Table 1 that Factors 6 and 7 admit of a single factor being rotated from them which loads the two Super Ego factors, G and Q_4; Desurgency, F(−); and positive performance on Reasoning, Fluency, Series, and Classification. It makes good psychological sense that such cognitive performances should be aided by such personality factors, but it suggests they are acting in the present test situation (as in Equation 1) rather than through help in past performance as in Equation 3.

caricature, too different for g_f ever to be identified with k. Finally, however, g_f is *not* particularly loaded on mechanical aptitude variables (the evidence is not given here, but in Horn, unpublished), or in spatial ability measures, or one-sidedly in any of those performances which have been included in this vague and impractical practical ability artifact.

On the last issue—that of giving us a far more complete descriptive differentiation of g_f and g_c in the extent and nature of their loading fields—there is admittedly a great deal still to be found out. Considering the volumes of ability correlating that have gone on over the past 20 years it has been, to say the least, theoretically uninspired. Especially the hypothesis should be tested that g_c's loading pattern will alter with age, subculture, and culture whereas g_f's will not. Already such studies as that of Sisk (1940) point to the effect on the form of the crystallized intelligence pattern of such very local influences as an "interest in study" over particular college courses shared by members of a group. However, one unjustified assumption needs to be scotched which has been, among teachers, as obstructive in holding up the practical introduction of culture-fair tests, as the k illusion has been in buttressing outworn theory. This is the "faith validity" judgment that because such tests employ shapes they must be testing spatial ability. As Figure 1 shows, the loading of the fluid general intelligence factor is not as great in Spatial as in Matrices, Topology, and Series tests, and though such tests depend on perception there is as yet no evidence that they load unduly Thurstone's perceptual factors.

If, as stated at the outset, the theoretical aim of culture-fair intelligence subtest construction is to require complex relation eduction in material that is either completely new to or completely overlearnt by all persons to be tested, it is of interest to discover what these areas may be. A systematic attack on this was begun some years ago in the work of Cattell (1940), Line (1931), Fortes (1931), and others and pointed to *simple* spatial perception, shade and pitch perception, and probably tactile experience and awareness of the body, as being sufficiently overlearnt in most cultures, but this exploration was never continued. Meanwhile, from a practical intelligence test construction standpoint it is interesting to note that the Classification, Topology and somewhat less, the Matrices tests are good measures of fluid ability (at any rate as used in the Culture Fair Test) and Number, Reasoning and Verbal primaries of crystallized ability. Incidentally, the evidence of cultural learning in the Series test is a bit surprising.

While commenting on odd loadings in Table 1 one may note also the loading which a factor, Number 8, outside the general ability factors, has on the Matrices test. The present writer has elsewhere argued, on the theoretical likelihood of such a loading, against the legitimacy of using Matrices alone, for measuring fluid general intelligence, instead of a complete Culture Fair battery averaging several subtests. This shows that indeed appreciable specifics or extra factors are as likely to occur in perceptual as in traditional subtests. It is possible they will not prove to be *cognitive* specifics, but, as the present analysis suggests, less than half of the variance in Matrices appears to lie in general intelligence, and the rest is sheer unknown bias in the test measurement if Matrices alone is used as an intelligence test.

As to the present accuracy of estimate of F_1 and F_2 the loadings of the subtests in the Culture Fair battery[18] on the fluid

[18] The search for further good subtests might now be guided by the emerging theory that

ability factor are, in general, as good as, or a little better than, those of the Thurstone primaries on the crystallized ability factor (mean=.40 as against .36).

NEEDED RESEARCH

At the present moment researches on these theories are as vital in the applied as in the basic field. For if the main concepts are correct, present routine testing practices in school, and still more in vocational selection of adults, are wasting a lot of talent. The traditional intelligence test, because it confounds fluid with crystallized intelligence and even with the third-order general attainment factor, gives a speciously good correlation, at least in schools, with the criterion. A test in June predicts academic achievement in June (or even in the following September) for the simple reason that it already surreptitiously contains in itself most of the criterion it is claiming to predict. In fact, if short-term correlation is all that is required it would be sensible to abolish intelligence tests and predict from today's school achievement to tomorrow's.

But the idea of introducing intelligence tests has surely much broader objectives. It aims to give psychological understanding of causes of backwardness or sources of high performance, to permit prediction of more remote performances in time and place, and to permit us to apply a psychological science via functional analytical concepts. The writer had a student who never went to school till 17, who was soon getting straight A's as a junior in college, he did splendidly on a culture-fair test at 17 but would have been excluded from college on his traditional intelligence test or achievement performance. The discrepancy of g_c and g_f is, however, mostly small in the school years, so long as we have homogeneity of school, cultural subgroup, and social status. It is in the adult field, e.g., in selecting for an intelligence-demanding learning opportunity among persons of different ages and distances from school activities, in different regions, that failure to measure g_f and g_c separately could lead to serious misdirection. How fair is the Miller Analogies to engineering students competing for graduate school positions against English majors, compared with a culture-fair test? It is well known that the norms of extensively standardized traditional intelligence tests are invalidated within 2 or 3 years of their production by changing (usually increased!) school and cultural education levels. Is this intergenerational "creep" of norms on crystallized intelligence tests avoided by culture-fair tests as preliminary data (Cattell, 1957b) indicates? These are the problems which applied research should be facing.

In terms of basic research a realm of fascinating concepts and possibilities is opened up. Is it possible that g_f minus g_c might be the best predictor of organic brain damage? Can g_f and g_c also be separated by intraindividual, *P* technique factoring, demonstrating the functional unity of each in terms of susceptibility to day-to-day changing conditions and stimuli? What clarity might

fluid ability shows itself most highly, relative to crystallized ability, in complex relation and correlate eduction performances where crystallized ability is eliminated either by (*a*) all subjects being entirely untrained in the performance, or (*b*) all subjects being completely overlearnt in the fundaments (but not the relations, else intelligence would disappear)! Possibly Culture Fair Subtest Number 4, Topology, is an example of the former and Spatial ability of the latter, since the latter is a common requirement of bodily movement for people in all cultures. This definition of one property of the fluid general ability needs to be experimentally pursued in relation to the other hypothesized properties above, now that the factor can be isolated.

be brought to the vexed question of preschool ability organization by these concepts? (There are important leads here in the work of Burt, 1955; Hebb, 1942; Hofstaetter, 1954; McNemar, 1940; Reitan, 1958.) Will nature-nurture ratios calculated on g_f measures prove to be altogether more highly genetically determined than g_c measures? Is the g_f IQ more constant?

Is it possible that what Thurstone called primary ability factors are psychologically the same as what we have called aids above? Probably both will consistently appear as first-order factors, relative to g_c and g_f as second order, and, although some may well be genetic in origin, the theory is worth exploring that many first orders will prove to be growths from a specific breakthrough (in the form of hitting upon a habit of analysis which becomes instrumental to a whole sequence of skill). In view of the elaborate writing of Piaget, and still more the speculations of his followers, regarding structures of this kind (see the recent summary by Hunt, 1961) it is high time that more precise correlational methods should attempt to demonstrate these patterns and relate them to the three types of formation here discussed.[14]

But perhaps good research strategy indicates that the first necessity today is the checking of the existence of the two general factors in groups of differing ages, homogeneities, and cultural settings, in relation to the obvious differences which would be expected from the theory. For example, this experiment needs repeating with age-homogeneous groups at 30 and 50 years of age; with adults heterogeneous as to age; and with mixed "normals" and known brain damaged cases. It needs to be tried with larger ranges of variables, identically used with two or three different cultural groups, to see whether the third-order achievement general factor can be separated both from the second-order crystallized intelligence and fluid intelligence factors, and especially we need age plots of the differing course of these two factors, each with and without speed limits in testing.

[14] In so far as the discovery of each aid (instrument for further problem solving) must be partly a function of fluid ability level, a part of the variance of all aid developments will be absorbed in the general factor of crystallized ability. The F_A's of Equation 3 would represent, therefore, only the narrow group factor variance left over after this absorption in the general factor.

SUMMARY

The theory of fluid and crystallized general intelligence factors has been examined in regard to an array of implications. A distinction has been drawn between a general achievement factor and the crystallized intelligence factor. Both, as distinct from fluid intelligence, will so change their pattern (*a*) with age, and (*b*) with culture as to constitute relatively poor predictors of other performances in groups of different ages and heterogeneous subcultures.

An experiment factoring 44 hypotheses-relevant variables measured on 277 seventh and eighth grade boys and girls, with sufficient noncognitive variables to permit effective rotation of any general cognitive factors which might appear, demonstrated the existence of two general ability factors. One fits the crystallized ability factor measured in traditional intelligence tests and the other a fluid general ability measured in culture-fair intelligence tests. Evidence is offered that the latter is neither a spatial ability factor nor the so-called "practical ability" (k) factor, but basic general intelligence. These two general abilities appear in a single third-order factor hypothesized to express the "formative fluid ability" partly responsible for the present level of both of them.

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